ORLANDO HEALTH
1414 Kuhl Ave.
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DEPARTMENT GUIDELINE

Guideline Title: **Cyanide Toxicity**	Guideline Number: **GUID-3203-T013**

Issue date: 09/09/2014	Replaces Dept. Policy:
Revision dates: 07/01/2026	Developed by: Air Care Team
Approved by: Dr. Danielle DiCesare, MD Air Care Team Medical Director	Approved by:
Signature:	Signature:
Department Numbers: 3203	

I. **PURPOSE:** Recognize and appropriately treat the victim of cyanide toxicity while providing supportive measures and rapid transport to definitive care.

PATHOPHYSIOLOGY:

Acute exposure to hydrogen cyanide causes significant morbidity and mortality in structure fires, household and industrial accidents, and suicides. Like carbon monoxide, cyanide is a colorless, odorless gas. This substance is found in fire smoke, particularly when synthetic materials such as plastics, acrylics, polyurethane foam, or insulation is burned. Thus, it is most encountered by exposure to household or industrial fires. However, not all patients who have suffered smoke inhalation from a closed space fire will have cyanide poisoning.

Hydrogen Cyanide impairs utilization of oxygen within the cells at the mitochondrial level. The signs and symptoms of cyanide exposure result from cellular hypoxia and anaerobic metabolism. These include shortness of breath, altered mental status, seizures, cardiac arrhythmias, and cardiovascular collapse. Patients with cyanide exposure may have an almond odor on the breath and will have significantly elevated lactate levels.

Success in treatment of cyanide poisoning is time critical with administration of an antidote soon after exposure potentially reducing morbidity and mortality. Given the lack of a rapid detection technique, cyanide toxicity must be treated based upon suspicion alone. Cyanide toxicity should be suspected in any victim of smoke inhalation who exhibits neurologic, respiratory, or cardiovascular compromise.

Hydroxocobalamin detoxifies cyanide by forming cyanocobalamin which is then excreted in urine. Several studies demonstrate the safety of this antidote making administration even in potentially nontoxic patients less of a concern than previously utilized antidotes

II. **DEFINITIONS:**

III. **DEPARTMENT GUIDELINE:**

a. Assess and manage airway, breathing, and circulation. Administer 100% oxygen regardless of pulse oximetry readings. Secure airway, if necessary, secure an advanced airway *per GUID3203-G008 Rapid Sequence Induction for Endotracheal Intubation* and *GUID3203-PR020 Endotracheal Intubation* and ventilate with a transport ventilator per *GUID3203-PR024 Ventilation with a Mechanical Ventilator*.

b. Assess risks for and signs of cyanide toxicity: Smoke inhalation or enclosed space fire, soot around mouth, nose, or back of mouth, abnormal ABCs, or altered mental status (GCS < 10 indicates severe poisoning)
c. IV access (draw blood tubes prior to administration of antidotes when possible)
d. If patient displays signs of inadequate perfusion, treat per *GUID3203-M011 - Hypotension*.
e. Consider Hydroxocobalamin/Cyanokit if clinical suspicion of Cyanide poisoning is high AND the patient presents with either altered mental status (GCS < 8), seizures, moderate/severe respiratory distress, unexplained hypotension, or cardiac arrest
    i. When reconstituting a CYANOKIT:
        1. Place vial in the upright position.
        2. Add 200 ml 0.9% NS to the vial using the transfer spike and fill to the line.
        3. The vial should be repeatedly inverted or rocked, not shaken for at least 60 seconds prior to infusion. Do not use if particulate matter is present or solution is not dark red.
        4. Use vented IV tubing.
    ii. Adult: **Hydroxocobalamin 5 grams IV/IO** over 15 minutes.
        1. If a favorable response is seen and a second dose becomes necessary or if severe symptoms persist, a second 5 gram dose may be considered at a rate range from 15 minutes (if in extremis) to 2 hours, as clinically indicated.
    iii. Pediatric: **Hydroxocobalamin 70 mg/kg IV** (maximum single dose 5 grams), may repeat once if needed.
    iv. Contraindicated in patients with known anaphylactic reactions to hydroxocobalamin or cyanocobalamin
    v. Infuse into a dedicated intravenous/intraosseous line.
    vi. Assess patient for redness or discoloration at the injection site, skin, or mucus membranes, hypertension, and flushing of the skin. Most common adverse reactions (> 5%) include transient chromaturia, erythema, oxalate crystals in urine, rash, increased blood pressure, nausea, headache, and infusion site reactions.
f. Treat seizure activity with benzodiazepines per *GUID3203-M015 – Seizures*.
g. Patients receiving Cyanokit should be transported to a burn center.

II. **DOCUMENTATION:** EMS Charts

III. **REFERENCES:**

a. www.cyanokit.com How to administer CYANOKIT.